Fraud Detection System using Isolation Forest and Gen AI

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Problem Statement

- Financial institutions process thousands of transactions daily.
- Manual fraud detection is time-consuming and error- prone.
- Traditional rule-based systems may miss unusual fraud patterns.
- An intelligent system is required to automatically identify suspicious transactions and provide explanations.

Project Overview

- **Data preprocessing**
- **Unsupervised Machine learning**
- **Generative AI**
- **Interactive Visualization**
- **Automated Report Generation**

The System Analyses transaction records, identifies anomalies using Isolation Forest, and generates contextual explanations using Gemini AI.

Tools Used

Component	Tool
Programming Language	Python
Data Processing	Pandas, NumPy
Machine learning	Scikit-Learn
Anomaly Detection	Isolation Forest
Visualization	Plotly
Generative AI	Gemini 2.5 flash
Development Environment	Jupyter Notebook

Dataset Description

The dataset contains banking transaction information such as:

- Transaction ID
- Account ID
- Transaction Amount
- Transaction Date
- Transaction Type
- Location
- Device Id
- Customer Age
- Occupation
- Login Attempts
- Account balance

The dataset was provided in excel format and was later processed and exported in CSV for Analysis.

Data Preprocessing

- **Step 1: Load Dataset**

The CSV file was loaded using python.

- **Step 2 : Data Cleaning**
- **Removed unnecessary columns**
- **Converted data types where required.**
- **Step 3: Feature Selection**

Important fraud- related attributes selected:

- **Transaction Amount**
- **Login Attempts**
- **Account Balance**
- **Customer Age**
- **Step 4: Safe transaction Filtering**

Basic filtering rules were applied to separate obvious normal transactions before anomaly detection.

Anomaly Detection Using Isolation Forest

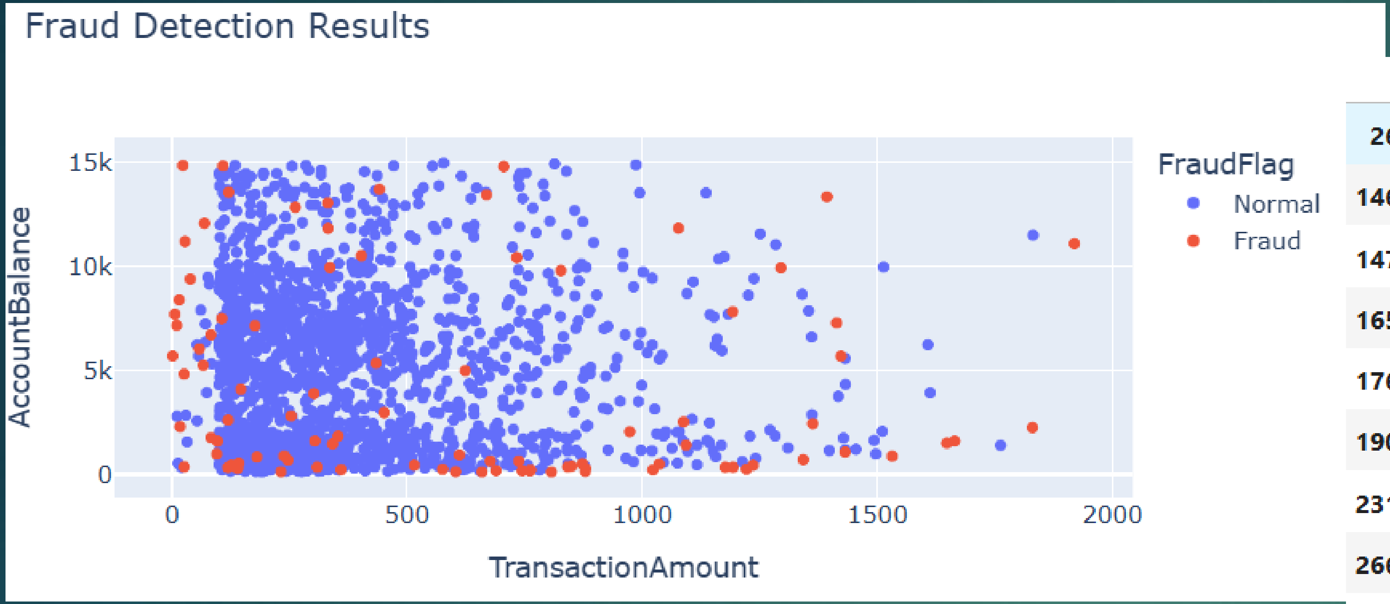
Isolation Forest is an unsupervised machine learning algorithm specifically designed for anomaly detection.

- **Model Workflow:**
- **Input transaction dataset**
- **Select numerical features**
- **Train Isolated forest Model**
- **Predict anomalies**
- **Mark suspicious as Fraud**
- **Generated explanations using Gemini AI**

	TransactionID	TransactionAmount	FraudFlag	AI_Explanation
26	TX000027	246.93	Fraud	Multiple login attempts
146	TX000147	973.39	Fraud	Unusually long transaction duration
147	TX000148	514.95	Fraud	Multiple login attempts
165	TX000166	231.91	Fraud	Unusual transaction pattern detected
176	TX000177	1362.55	Fraud	High transaction amount
190	TX000191	1422.55	Fraud	High transaction amount
231	TX000232	705.60	Fraud	Unusual transaction pattern detected
266	TX000267	66.90	Fraud	Multiple login attempts
274	TX000275	1176.28	Fraud	High transaction amount, Multiple login attempts
291	TX000292	1036.21	Fraud	High transaction amount

```
FraudFlag
Normal    1723
Fraud       91
Name: count, dtype: int64
```

Fraud Detections Results



	TransactionID	AccountID	TransactionAmount	Location	Channel	FraudFlag
26	TX000027	AC00441	246.93	Miami	ATM	Fraud
146	TX000147	AC00385	973.39	Sacramento	Branch	Fraud
147	TX000148	AC00161	514.95	New York	Online	Fraud
165	TX000166	AC00182	231.91	Memphis	ATM	Fraud
176	TX000177	AC00363	1362.55	El Paso	ATM	Fraud
190	TX000191	AC00396	1422.55	Washington	Branch	Fraud
231	TX000232	AC00430	705.60	Phoenix	Branch	Fraud
266	TX000267	AC00178	66.90	Charlotte	Online	Fraud
274	TX000275	AC00454	1176.28	Kansas City	ATM	Fraud
291	TX000292	AC00284	1036.21	Omaha	Online	Fraud

Key Observations

- Isolation forest successfully identified anomalous transactions.
- Fraudulent transactions showed unusual transaction patterns.
- High login attempts and abnormal transaction amounts were common indicators.
- Gen AI provided human – readable explanations for flagged transactions.

Gen AI-Powered Explanation System

To improve interpretability , Gemini AI was integrated.

The model generates explanations for suspicious transactions based on:

```
print("Total Transactions:", len(df))
print("Safe Transactions:", len(safe_transactions))
print("Transactions Checked:", len(suspicious_pool))
print("Fraud Detected:", len(fraud_transactions))
```

```
Total Transactions: 2512
Safe Transactions: 698
Transactions Checked: 1814
Fraud Detected: 91
```

Interactive dashboard

The dashboard was made using Jupyter notebook

Features:

- **Fraud transactions display**
- **AI generated explanations**
- **Interactive Visualizations**
- **Search and filtering capability**
- **Export functionality**

Workflow Automation

The workflow is automated through Python scripts:

- **Load transaction data.**
- **Perform Preprocessing**
- **Run Isolation Forest**
- **Detect Anomalies**
- **Generate AI explanations**
- **Export results to CSV**

Output file:

Fraud_Detection_Final_Dataset.csv

Testing and Validation

Validation steps:

- Verified anomaly detection results.
- Checked generated AI explanations.
- Confirmed CSV export functionality.
- Verified dashboard visualizations

Conclusion

This project successfully demonstrates an end to end fraud detection pipeline using machine Learning And Generative AI.

The solution:

- **Detects suspicious transaction.**
- **Generates contextual explanations.**
- **Provides visual analytics.**
- **Automated fraud reporting.**

Future improvements may include real-time fraud monitoring, emails and deployment as a web application.